

Module 04: Cognition — Language Models and Reasoning

Learning Objectives

1. Explain how a **Large Language Model (LLM)** works: predicting the next token given a sequence of preceding tokens.
2. Distinguish between **narrow AI** (specialized) and **broad/general AI** (flexible).
3. Evaluate claims about machine "understanding" versus sophisticated pattern matching.

Introduction

Can a computer think? When ChatGPT writes an essay, is it "understanding" the topic, or is it performing incredibly sophisticated pattern matching? This module explores machine cognition — what AI systems can and cannot do — and connects it to the Active Inference view of cognition as generative model inference.

Key Concepts

1. How Language Models Work

A Large Language Model (LLM) like GPT works by predicting the next word (token) in a sequence:

Given: "The cat sat on the ____" Prediction: $P(\text{"mat"}) = 0.35$, $P(\text{"floor"}) = 0.20$, $P(\text{"table"}) = 0.15$, ...

The model was trained on trillions of words, learning the statistical patterns of human language. In Active Inference terms, the LLM is a massive generative model that has learned to predict human language with high precision. Each generated word reduces the "surprise" of the text.

2. Narrow AI vs. General AI

Feature	Narrow AI	General AI (AGI)
Scope	One task (chess, image labeling, translation)	Any intellectual task
Current Status	Exists and is highly capable	Does not yet exist
Active Inference View	Fixed generative model for one domain	Flexible, hierarchical generative model that transfers across domains

3. The "Understanding" Debate

Does GPT *understand* what it writes? Two perspectives:

- **Strong AI** position: If the behavior is indistinguishable from understanding, it *is* understanding.
- **Weak AI** position: Understanding requires consciousness, embodiment, or causal reasoning that statistical pattern matching cannot provide.

Active Inference contributes a nuanced view: a system that has built a generative model of a domain is *functionally* making inferences about that domain. Whether this constitutes "real" understanding may depend on the richness and flexibility of the generative model.

Applications

- **Autocomplete**: When your phone suggests the next word in a text message, it is running a small language model that minimizes prediction error based on your personal typing history.
- **AI-Generated Art**: Image generation tools (DALL-E, Midjourney) have generative models trained on millions of image-text pairs. They "imagine" images by inverting the prediction process: generate pixels that minimize the prediction error between the generated image and the text prompt.

Discussion Questions

1. If you gave GPT a physics problem it had never seen before and it solved it correctly, would that constitute "understanding"? What additional test would you need?
2. An LLM can generate convincing medical advice but has no body and no experience of being sick. Is this dangerous? Why?

Summary

LLMs are large-scale generative models that predict language with high precision. Whether this constitutes "understanding" remains an open philosophical question. Active Inference frames cognition as generative model inference — a lens that applies to both biological brains and artificial systems.

References

- Vaswani, A. et al. (2017). Attention Is All You Need. *NeurIPS*.